Big Data Application in Business

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OUTLINE

- Introduction to Big Data
- Big data = Competitiveness
- Big Data's application and prospect

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What is Big Data? (1/2)

- Assumption
  - 1/5 world's people will use the Internet
  - half of them are used to Google search
  - a person search twice in a day

Total search volume = 7B (persons) * 1/5 * 1/2 * 2 (times/person)

= 1.4B times = $1.4 * 10^9$ times

if the size of a search transaction = 1KB

$\Rightarrow 1.4 * 10^9$ KB ~ 1 TB

What is Big Data? (2/2)

- Big Data is a **huge volume** of information, which is structured in different ways.
- These volumes are so **extensive** and complex that a ***traditional approach does not manage to operate*** with it.
- That is why the information can **be only processed with high-load systems**, i.e. machine learning or other advanced technologies.

Data Engineering collect and validate large sets of information	**Data Storage** keep large data sets and enable the real-time operations control	**Data Analytics** examine varied data sets and provide information for crucial business decision
ETL Tools or Extract Transform Load, Apache Kafka, Python, R	Distributed file systems, NoSQL and in-memory databases	Apache Spark, Graph analytics, machine learning, Stream analytics

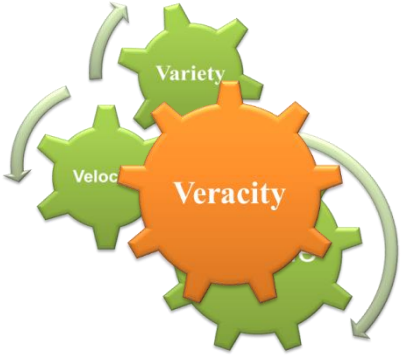
Large Data vs. Big Data

- There is a lot of data, you can only call it large data, not big data.
- Big Data is very big in volume, high at velocity and various types.
- Data must be used and analyzed effectively to be considered big data.

Traditional Data vs. Big DATA

	Traditional Data	Big Data
Structure of data	Structures are defined	Mix of Structured, semi-structured and unstructured data
Data volume	Based on business volumes and extent of digitation	Very high, in terabytes or petabyte and even more
Variety of data sources	Data source from database systems	Besides data from business information systems, text (email, documents), weblogs, sensors, RFID, etc.
Velocity	Low to moderate based on volume of business	High velocity
Sources of data	Organizational data, trading partners data	Organizational data, RFID, sensor data, Google searches, social media (FB, Twitter, Linked in, etc.)
Analytics	Provided historical view, status report	Real-time, direct feedback from consumer, sentiment analysis, opinions
Functions	Advises senior executives on internal business decision, focused on analyzing data for internal business decisions	Customer facing functions get direct market feedback which can be used for planning market strategies, planning etc., focused on customer, products and services

4V of Big Data



- Volume (amount of data) 巨量性
- Velocity (speed of data in and out) 即時性
  - Update fast
  - Time-sensitive, should be processed quickly
- Variety (range of data types and sources) 多樣性
  - Multiple data sources
  - Various data types (structured, semi-structured
- Veracity (trustworthiness of the data) 真實性
  - Data authenticity is the key to creating data

Critical Poin

Applying ar siness decisions ar tion.

VALUE

大數據特性: 大、快、雜、疑

Big data is getting more important.

- A lot of data is generated every day.
- Digitalized data is easier to analyze.

Social Site

Weather forecast

Stock Analysis

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Changes Caused by Big Data.

Quantitative changes cause qualitative changes.
More large and more diverse but more useful!

Who Can Apply Big Data?

- Have large amounts of data
- Can discover the information in large data
- Know how to turn information into intelligence.
- Can use this information to monetize

Who Can Apply Big Data?

- ~~Have large amounts of data~~
- ~~Can discover the information in large data~~
- Know how to turn information into intelligence.
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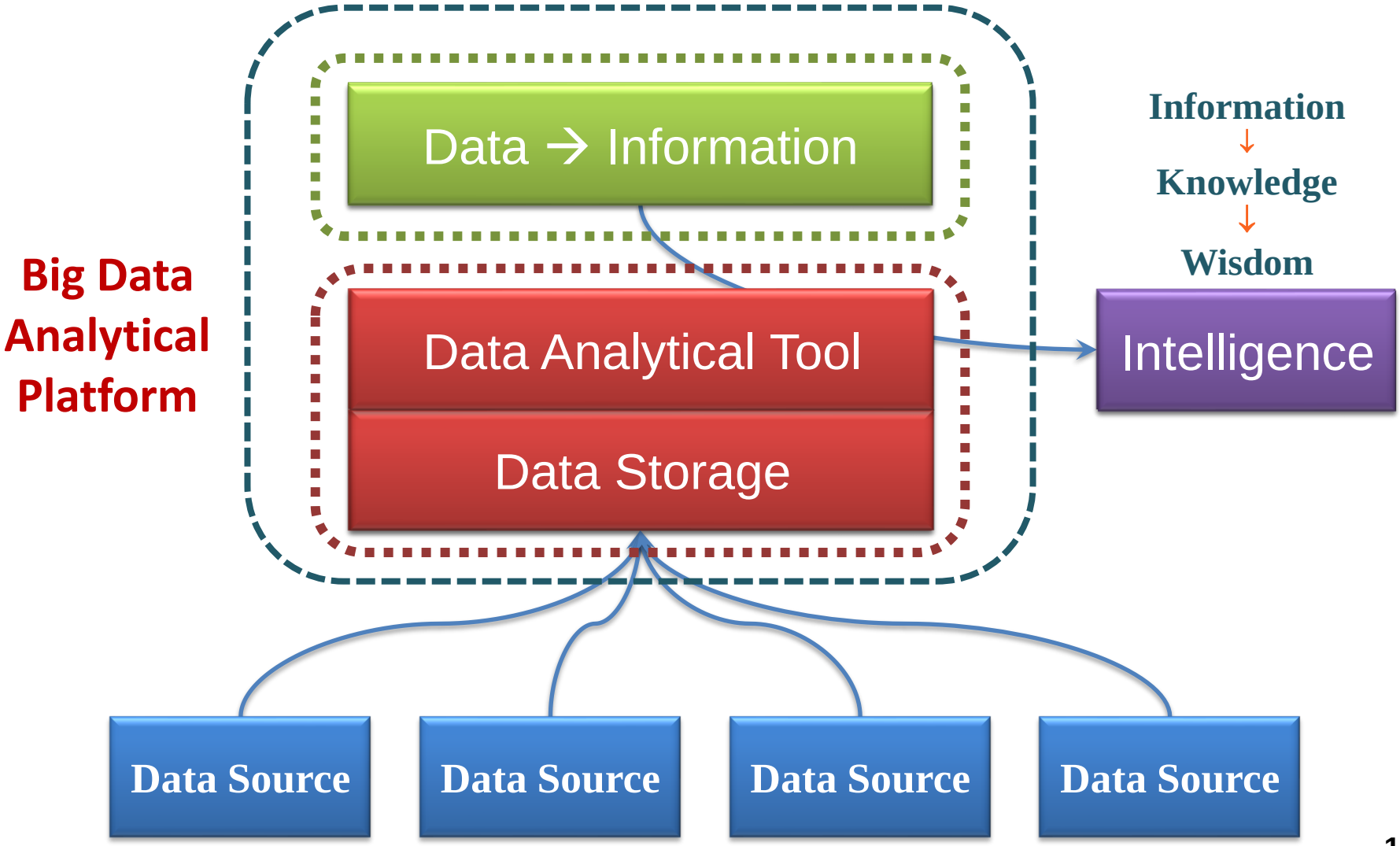
Levels of Big Data

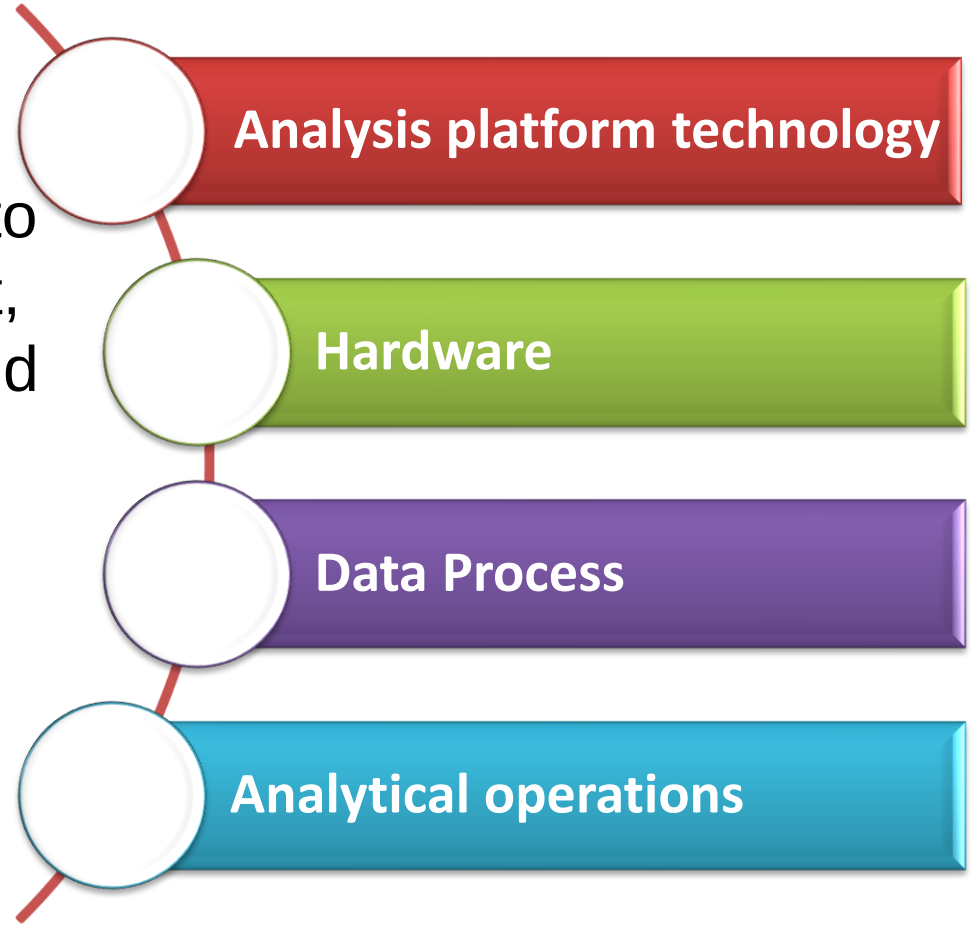
- What can be saved, is storage.
- What you can see, is data.
- Understandable, is called information.
- Only can be applied, is called intelligence.

System Architecture of Big Data



Big Data Analytics

- Use powerful **new information technology** to effectively collect, transmit, store, process, analyze and interpret all kinds of information presented by huge amounts of data, including the **correlation**, **causality**, **identification**, **trend observation**, etc..



The core requirements of BDAP

Can process structured & unstructured data

- Ability to quickly load and process raw multi-structured data
- Ability to store, process, and analyze unstructured data
- Can continue to load-> transformation -> processing-> load ...

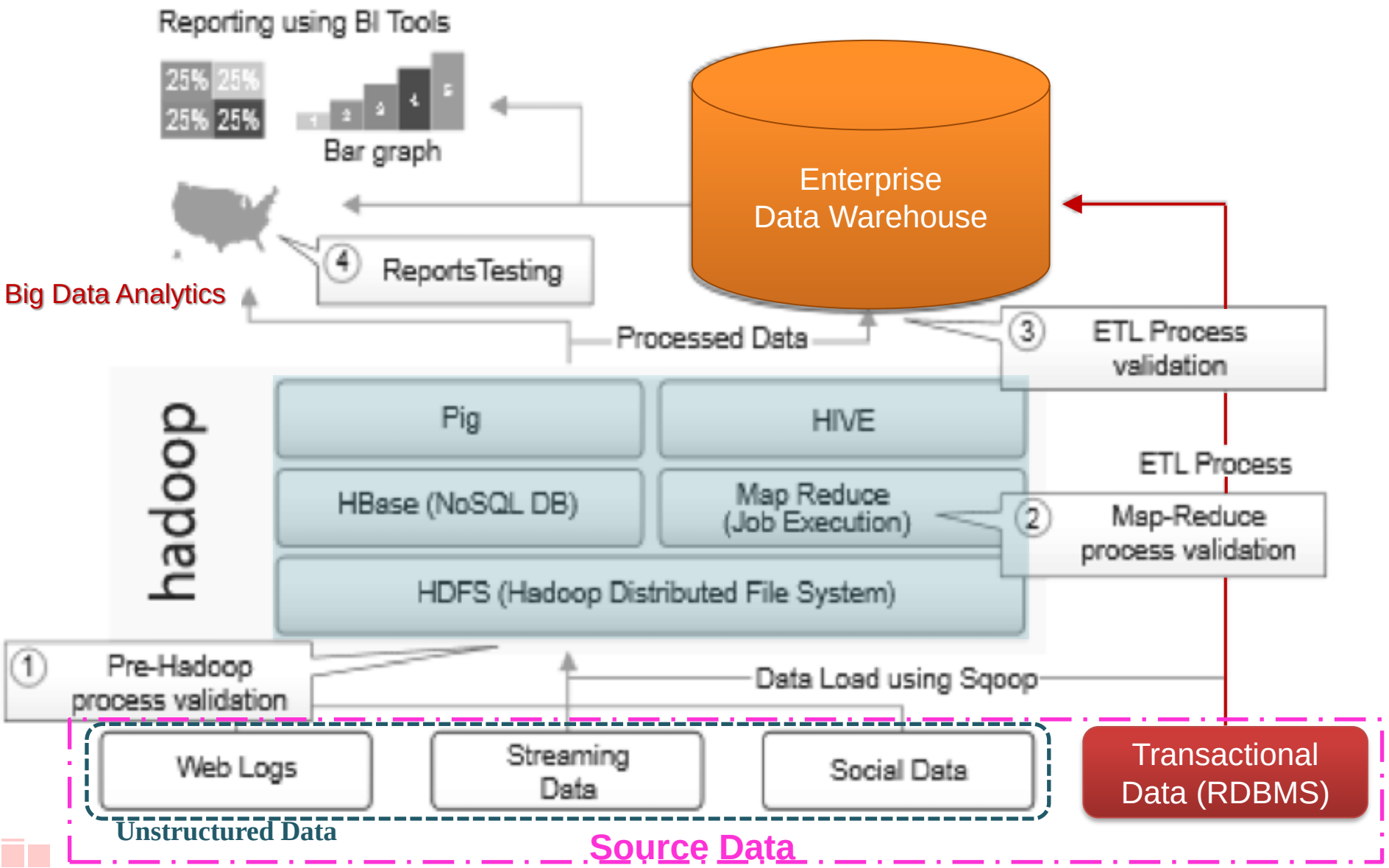
Ability to support various analysis tools

- ANSI SQL, reporting tools, ETL tools
- MapReduce (e.g. Hadoop)
- In-DB data analytics language (e.g. R / Python), in-DB data mining tools (e.g. SAS / Clementine)

Staff & Efficient hardware

- Data scientists can use programming (Java / C ++ / Python / Perl), statistics and SQL on it.
- Efficient analysis efficiency
- Can be easily expanded from TB to PB at a reasonable cost.

Example of BDAP



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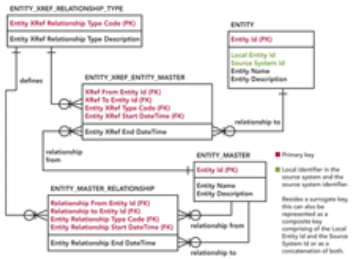
Industrial application trends

- Integrate with enterprise service platforms, to build analysis systems to obtain business information and support enterprise decision-making, thereby enhancing enterprise competitiveness and creating new enterprise value.

Data Extraction	Data Processing	Data Analysis	Value Added
Information mastery	Powerful Calculation	Expertise Consolidation	Utilization results for business value added
例: facebook, telecom …	例: Teradata, IBM, …..	例: Banking, Retail, Telco...	例: App Vendor, …..



Big Data Applications in Industry

- Product recommendation
- Data Monetization
- Extend business scope by using big data

e-Commerce

Finance

- Precision Marketing
- Risk management
- Business Improvement
- Service Innovation
- Product Innovation

- Network intelligence improvement
- Precise discovery of Customer needs
- Information Security Management
- New Services/products Development

Telecom

Retial

- customer profiling
- Social media analysis
- Dynamic pricing

Medical

- Remote monitoring
- Chronic disease management
- Cancer prediction

Big Data Applications

Retail

- Though data analysis, Japan 7-11 found the significant and high correlation between temperature and best-selling items.
  - The temperature difference in a day >= 7 ℃, the temperature difference between 2days >= 5 ℃ , and the humidity difference >= 30%
    - People who have a cold will increase.
    - Put cold medicines, thermometers and masks on the shelves.
  - Temperature between 24 ℃ and 27 ℃
    - Eel, ice and sunscreen will hot sell
  - Temperature between 23 ℃ and 25 ℃
    - Cold drinks, ice coffee and pesticides are essential.

Big Data Applications

e-Commerce

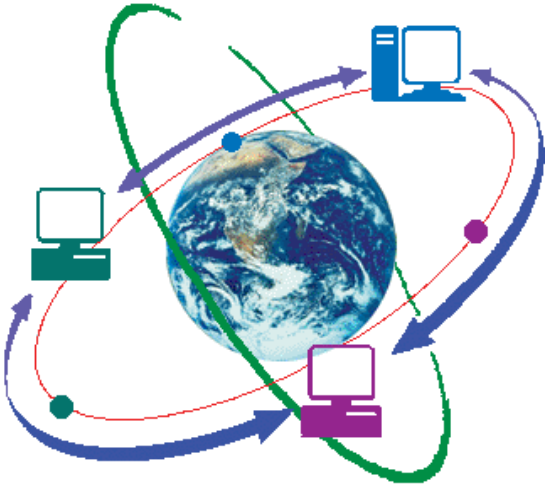
- Amazon-“Sending goods in advance”

  In order to shorten the logistics time, Amazon shipped out before placing an order. At first, due to US is vast place, the average delivery time is 15 days from order to receipt. Long delivery will reduce the online shopping willingness of customers. Now the service of sending goods in advance is greatly enhances people's willingness to shop online

Big Data Applications

Entertainment

- In 2030, broadcast TV will be completely replaced by Internet TV (Netflix, 2013).
- 80% of users' choice of video content will be affected by big data analysis recommendations.
- The big data application subverts the traditional video industry. By analyzing user preferences and personalized recommendations can give users different types of video content to their personalized user interface.
- Based on personal interests to customize video thumbnail.
- Using big data analysis results to make homemade popular albums (e.g. House of Cards)
  - Through the analysis of the ratings of all subscribed viewers, Netflix knows what kind of themes can attract what kind of viewers, and launches more favorite programs for different viewers to increase the click rate.



DATA APPLICATION

Passenger flow analysis

Location recommendation

Best Location Suggestion (1/2)

- **Business Question**
  - store location and operational hours
  - location with maximize passenger
  - how many similar business stores in area
  - best operational time slot
  - passenger profile

and customers

meal time
cations
oloration

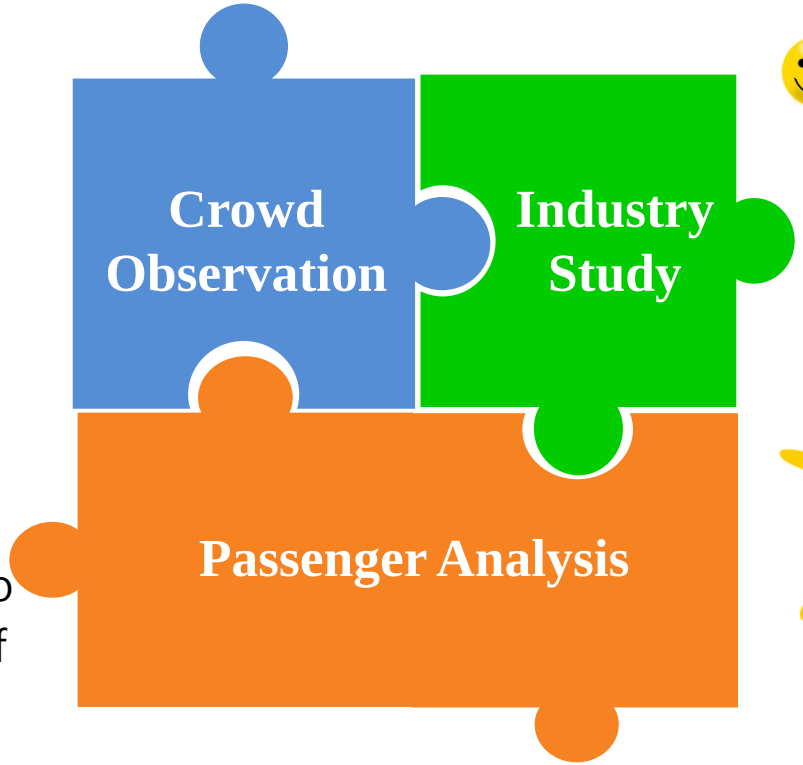
Best Location Suggestion (2/2)

Crowd Observation

- Analysis by meal time
- Analysis by weekday & weekend

Passenger Analysis

- Gender ratio
- Age distribution
- officer vs. residents ratio
- Customer proportion of web browsing preferences



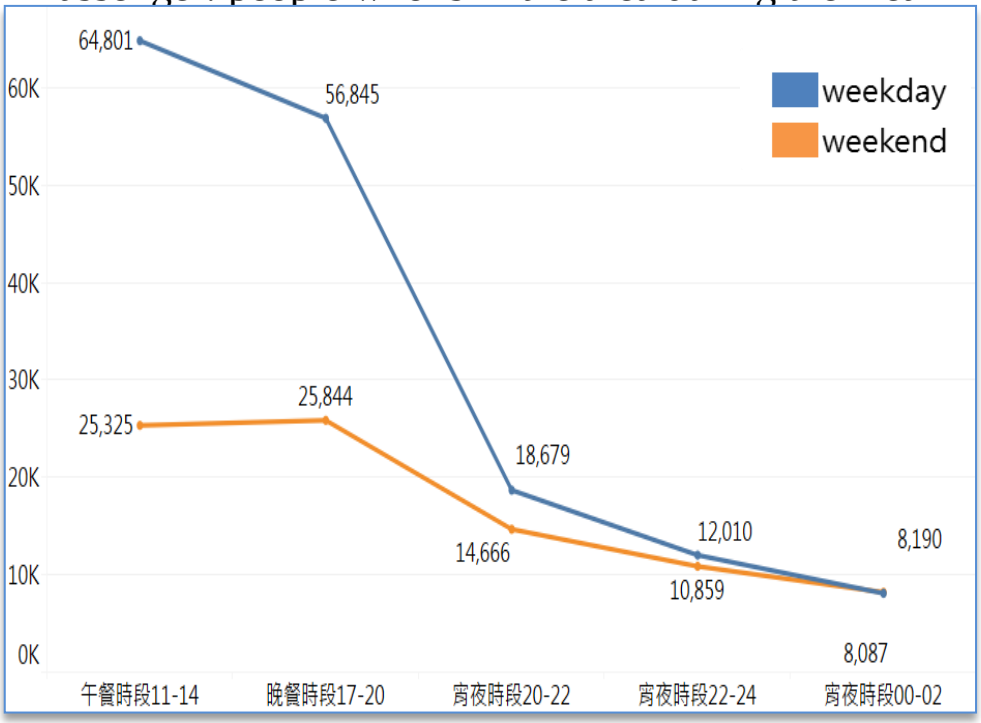
Industry Study

- Industry distribution
- # of the specific industries & the WOM Score of each store

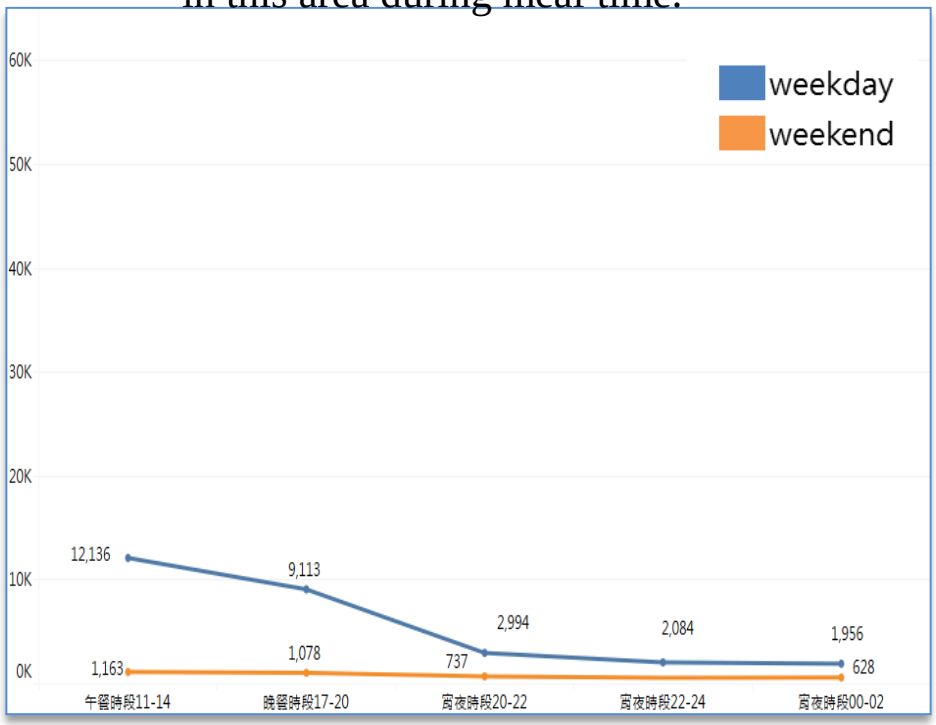
Platform with passenger flow and Google merchant data integration can provide users with accurate business decisions.

Crowds Observation-Meal Time

Passenger: people who is in the area during the meal



Meal: people who stay longer than 30 minutes in this area during meal time.

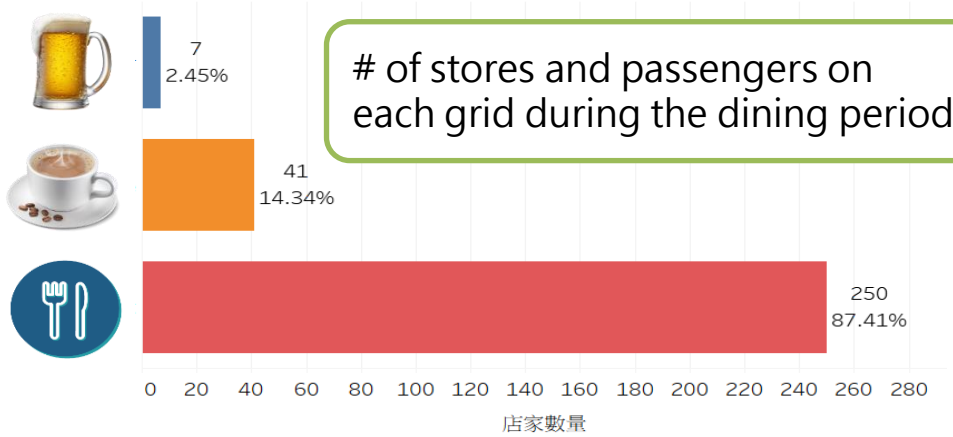


- The later the meal time, the fewer the crowd.
- The crowds drop suddenly after dinner.

- The crowds of meal time is very different between weekday and weekend.
- There is almost no crowds at the weekend.

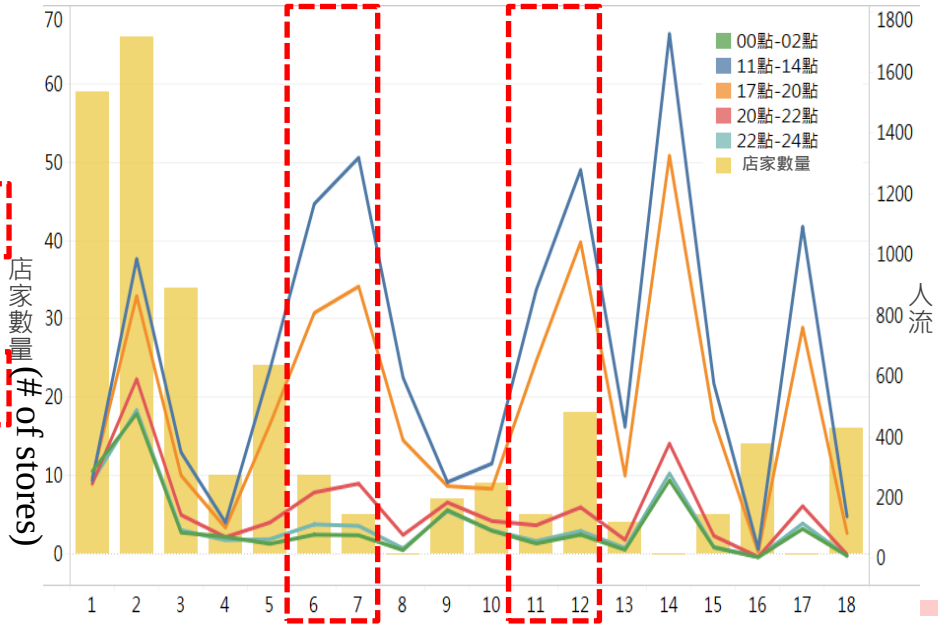
Industry Study

Related Industry Analysis



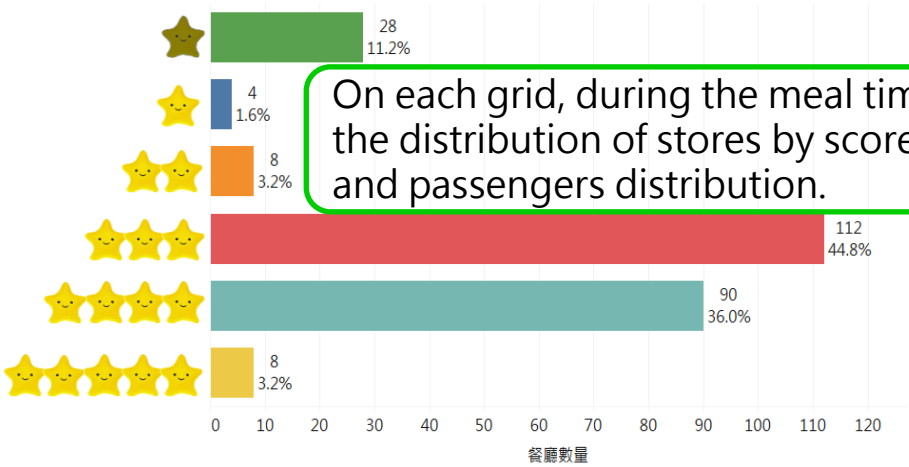
# of stores and passengers on each grid during the dining period

網格	總店數	餐廳	咖啡廳	酒吧	11-14點人流	17-20點人流	20-22點人流	22-24點人流	00-02點人流
1	59	57	1	1	258	267	246	254	287
2	66	61	6	1	986	864	590	489	476
3	34	28	6	0	349	273	143	93	85
4	10	9	1	0	118	101	70	60	71
5	24	19	8	0	621	441	118	63	48
6	10	9	1	0	1,167	808	217	112	78
7	5	4	2	1	1,319	895	246	107	76
8	0	0	0	0	595	389	77	33	27
9	7	5	3	0	251	238	184	155	159
10	9	6	4	0	312	229	123	92	91
11	5	4	1	0	884	650	109	57	49
12	18	14	4	0	1,279	1,041	168	90	78
13	4	4	0	0	433	272	61	34	28
14	0	0	0	0	1,727	1,326	378	280	257
15	5	3	0	3	576	456	74	38	36
16	14	13	2	1	31	20	5	4	4
17	0	0	0	0	1,092	760	172	115	97
18	16	14	2	0	138	83	13	8	9

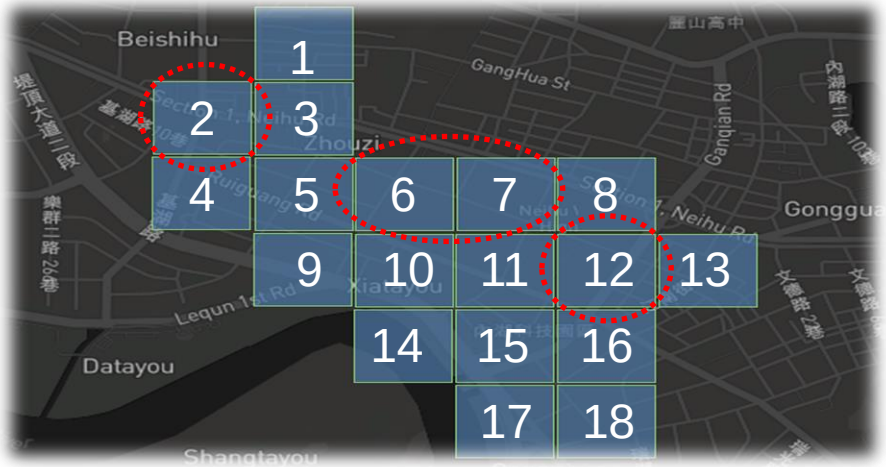


Industry Study

competitors vs. passenger flow



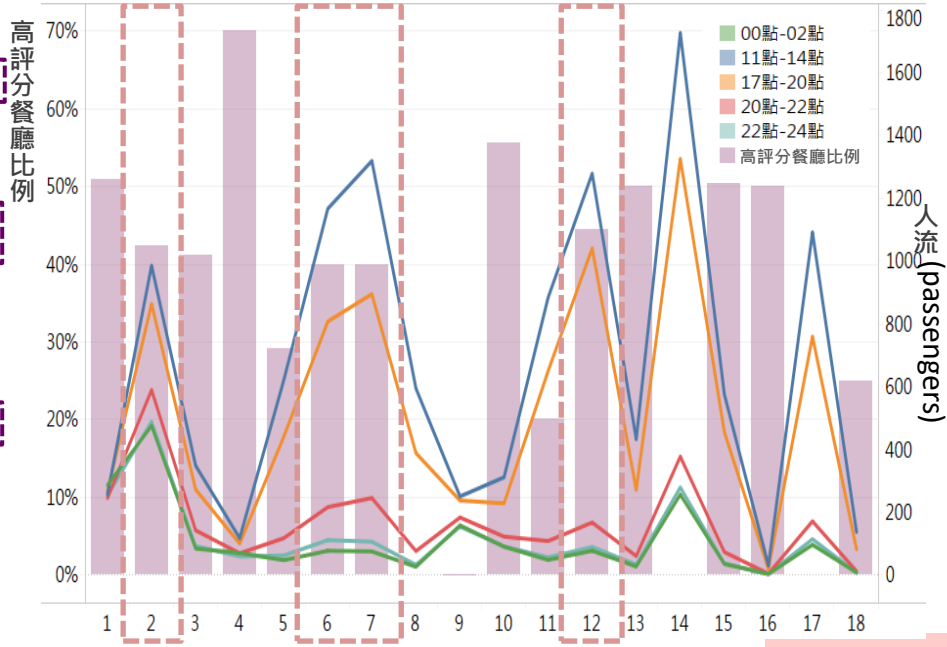
On each grid, during the meal time, the distribution of stores by score and passengers distribution.



網格	無評分	1分以上	2分以上	3分以上	4分以上	5分	高評分餐..	11-14點..	17-20點..	20-22點..	22-24點..	00-02點..
1	6	2	3	18	24	6	51%	258	267	246	254	287
2	7	1	1	29	28	0	42%	986	864	590	489	476
3	4	0	1	15	12	2	41%	349	273	143	93	85
4	1	0	0	2	6	1	70%	118	101	70	60	71
5	4	0	0	13	6	1	29%	621	441	118	63	48
6	1	0	0	5	3	1	40%	1,167	808	217	112	78
7	0	1	0	2	2	0	40%	1,319	895	246	107	76
8	0	0	0	0	0	0	無	595	389	77	33	27
9	0	0	1	6	0	0	0%	251	238	184	155	159
10	1	0	1	2	5	0	56%	312	229	123	92	91
11	1	0	0	3	1	0	20%	884	650	109	57	49
12	3	0	0	7	7	1	44%	1,279	1,041	168	90	78
13	0	0	0	2	2	0	50%	433	272	61	34	28
14	0	0	0	0	0	0	無	1,727	1,326	378	280	257
15	0	0	1	1	3	0	60%	576	456	74	38	36
16	1	0	0	6	7	0	50%	31	20	5	4	4
17	0	0	0	0	0	0	無	1,092	760	172	115	97
18	4	0	0	8	3	1	25%	138	83	13	8	9

高評分餐廳比例

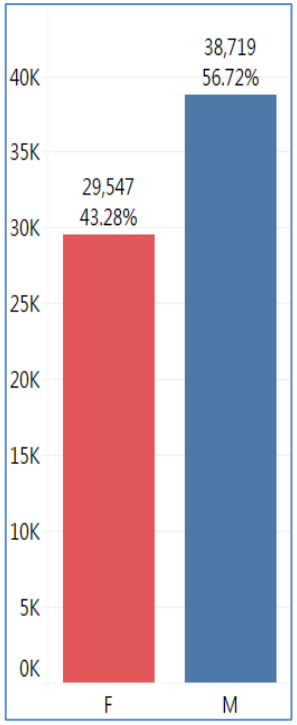
0% 70%



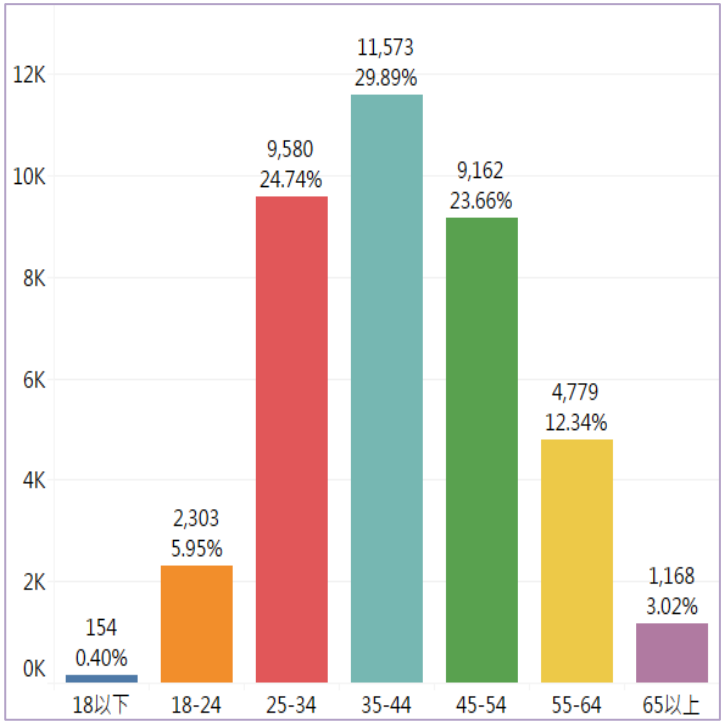
Passenger Analysis

gender、age

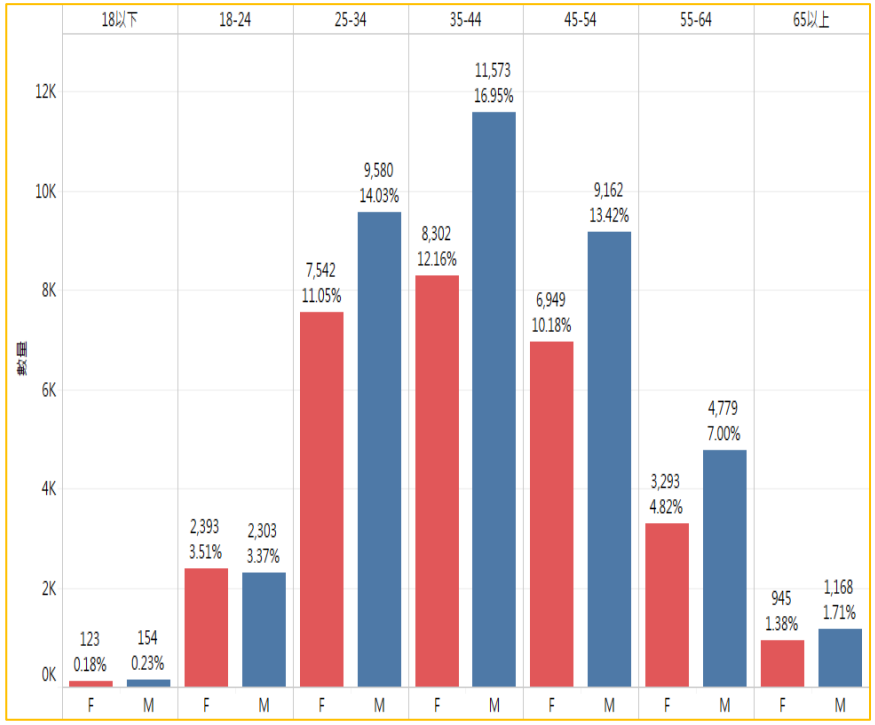
Gender



Age Segment

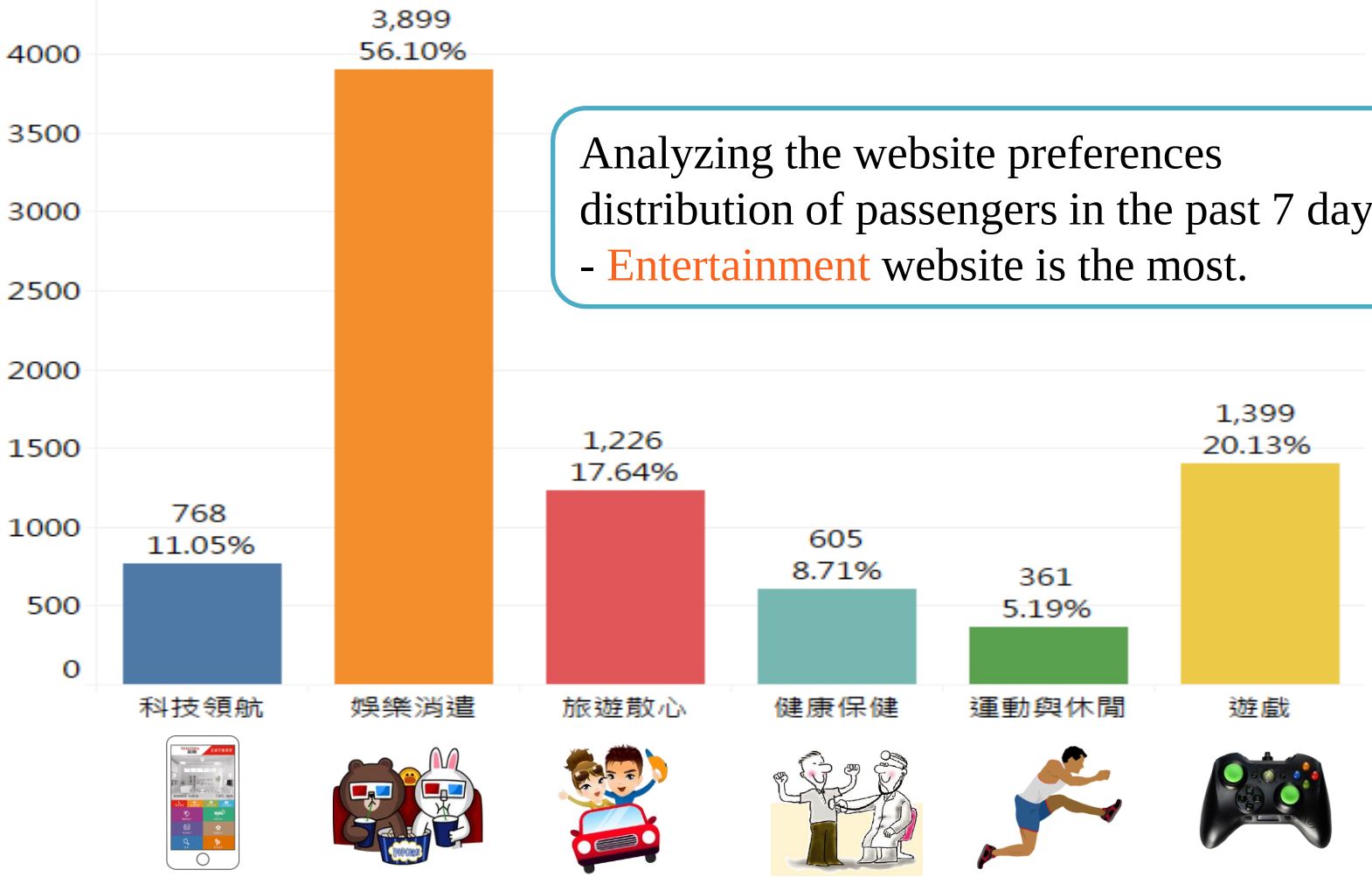


Gender vs. Age Segment



- Most passengers are between 25-54 years old.
- # of Male > # of female.

Passenger Analysis
websites preferences



Analyzing the website preferences distribution of passengers in the past 7 days.
- Entertainment website is the most.

Best Location

Crowd Observation

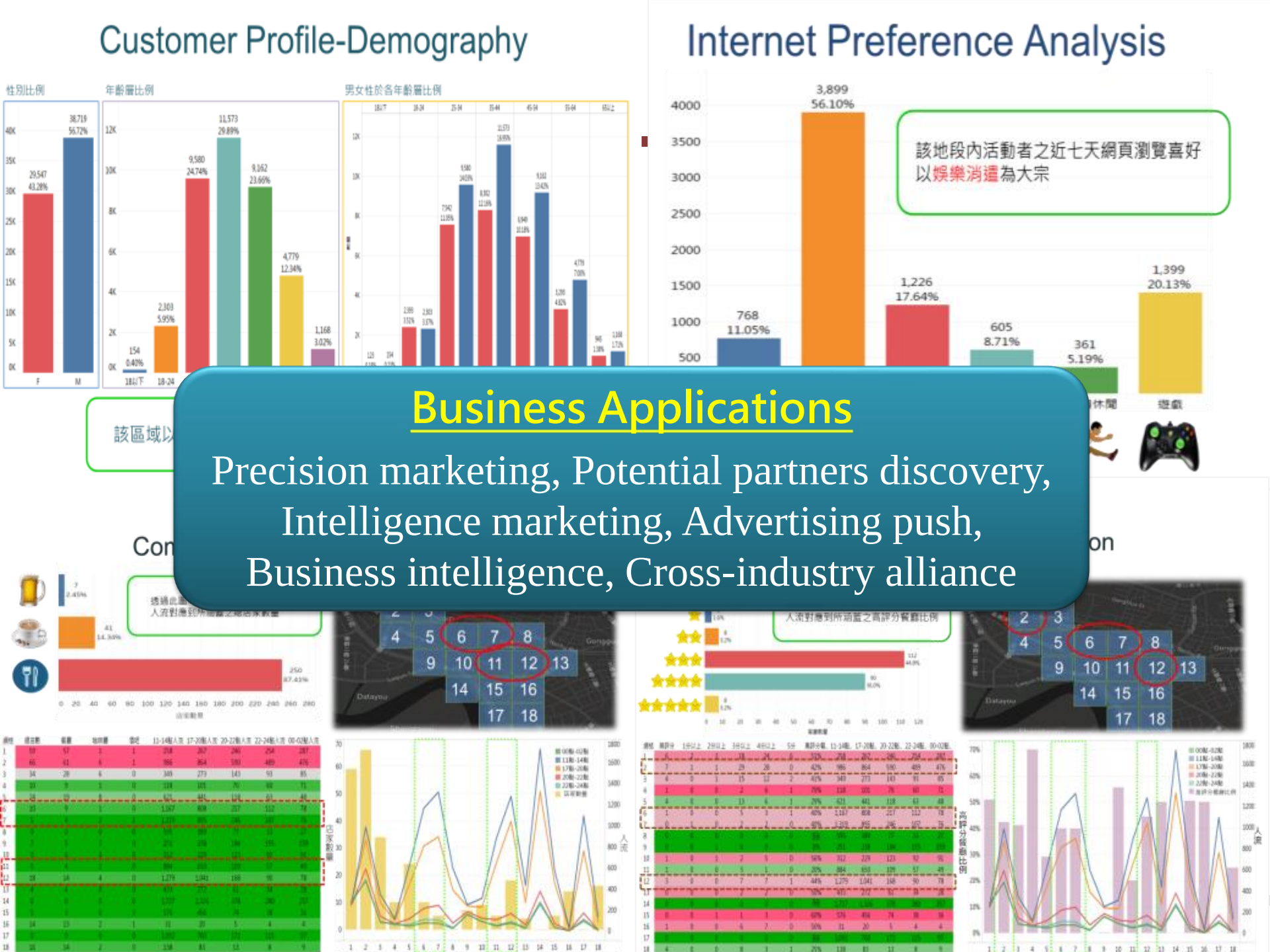
- Lunch time is the peak of people flow on weekdays.
- No one showed up on weekend.

Passenger Analysis

- Age 33-44 is the major customers & male > female.
- Most of these people like to browse Entertainment websites.

Industry Study

- Areas 6, 7, and 12, with high passengers flow and more stores of high-rating score.
- These areas are the best locations for business.


Customer Profile-Demography
Internet Preference Analysis
該地段內活動者之近七天網頁瀏覽喜好以娛樂消遣為大宗
3,899 56.10%
768 11.05%
1,226 17.64%
605 8.71%
361 5.19%
1,399 20.13%
Business Applications
Precision marketing, Potential partners discovery,
Intelligence marketing, Advertising push,
Business intelligence, Cross-industry alliance

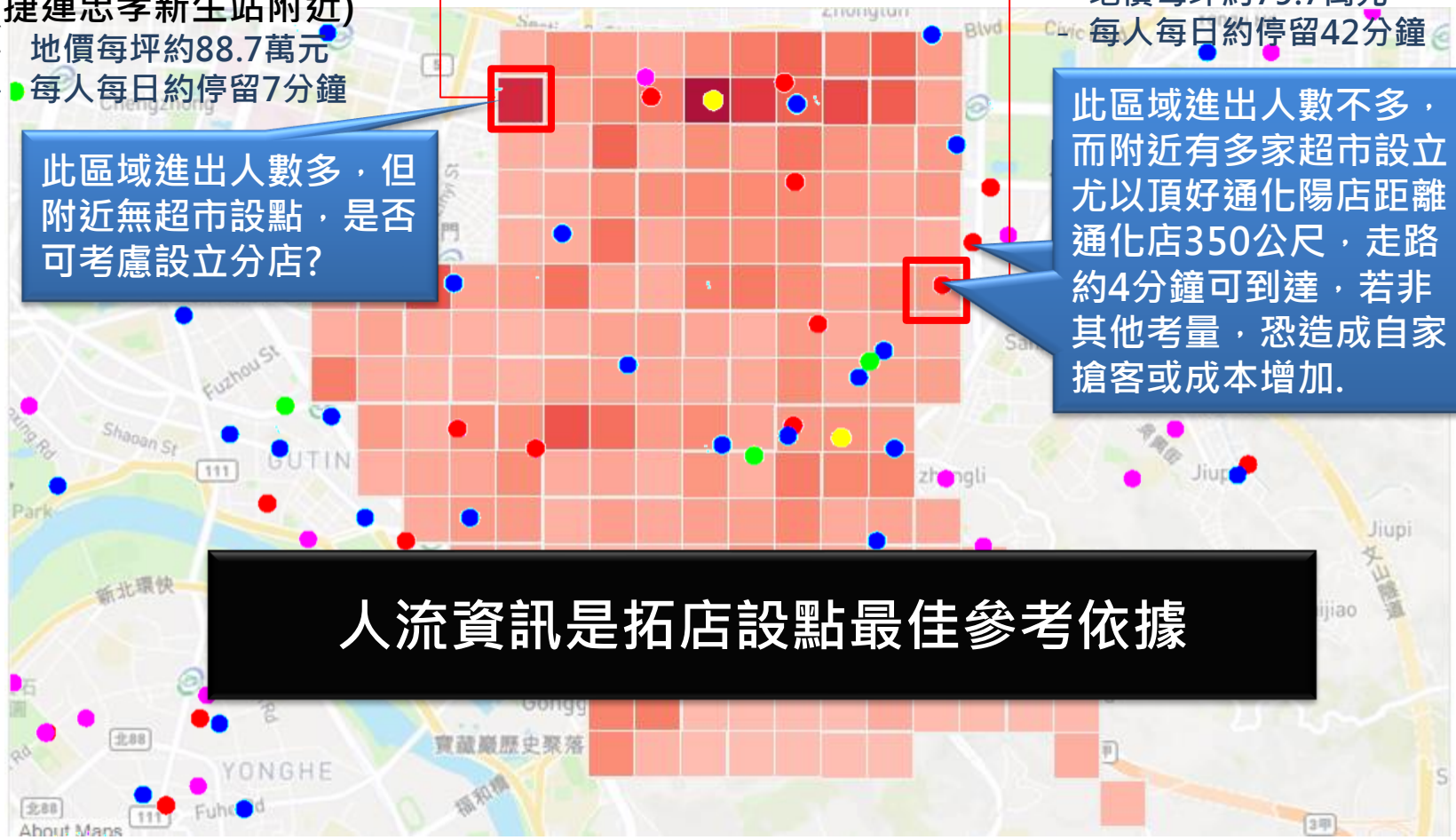
Best Location Selection

Information of People Flow

以大安區為例

大安區民輝里每日
約 6.4萬人進出
(捷運忠孝新生站附近)
- 地價每坪約88.7萬元
- 每人每日約停留7分鐘

大安區通化里每日約 1.7 萬人進出
(捷運信義安和站附近)
- 地價每坪約79.7萬元
- 每人每日約停留42分鐘



Big Data Used in Daily Life

- Big data is not only used in enterprises, but also often used in daily life.

Big data can also track and analyze the situation of one mo sug bod qua

Through the prediction and calculation of big tors d s.

The wonderful use of big data is to process data, interpret data, and make predictions in a scientific way.

By wearing measuring equipment, athletes can understand sports performance, posture and angle, and then analyze various improvement suggestions to improve sports performance.

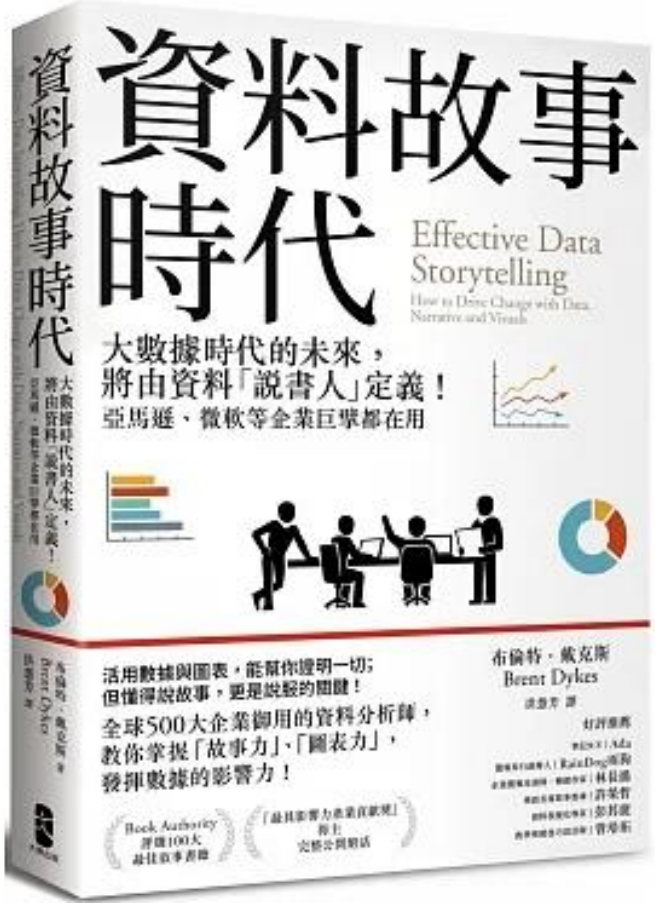
Sport Performance

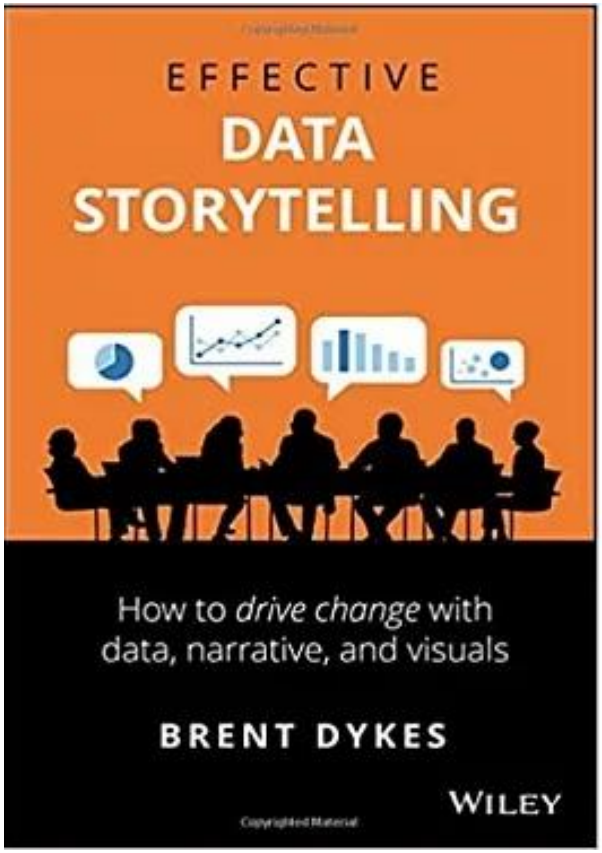
Security & Law Enforcement

United States uses big data to prevent or respond to terrorism and issues such as fraudulent transactions.

Effective Data Storytelling

A great data storyteller is one who can transform his insights and data visualizations into appealing, impactful data stories.





thank you
grazie
gracias
merci
danke
arigato
efharisto
spasiba
hvala
cheers
kiitos
waita
madlobt
aabar
zikomo
salamat
nkosi
tenki
danki
dankewol
matondo
ahsante
grassie
modupe
dziakuju
blagodaria
dziekuje
marahaba
shukran
buznyg
dèkoju
meharbani
dankegon
mèsi
manjuthe
taiku
sulpáy
saha
tack
waybale
blagodaram
maururu
bayarlalaa
obrigada
ngiyabonga
multumesc
tanemirt
shukriyaa
köszönöm
murakoze
gratzias
kinisou
merkzi
tanmirt
nuhun
akiba
dekuji
wado
barkal
mahalo
tänan
nandri
aitäh
menlau
soolong
manana
spas
saðol
gràcies
stuutiyi
shukria
skee
paldies
dakujem
bedankt
talofa
vinaka
rahmat
supas
sobodi
dankie
omol
shakkran
misaotra
chokrane
takk
trugéré
welalin
tashakor
matondi
obrigado
miigwech
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